UNION PUBLIC SERVICE COMMISSION (UPSC)

CDS GENERAL KNOWLEDGE — SUBJECT 3: GEOGRAPHY (PHYSICAL, INDIAN & WORLD)

OFFICIAL MASTER ANSWER KEY & COMPREHENSIVE EXPLANATIONS

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	(A)	7	(A)	13	(A)	19	(A)	25	(A)	31	(D)	37	(A)		
2	(C)	8	(C)	14	(C)	20	(A)	26	(A)	32	(C)	38	(A)		
3	(C)	9	(A)	15	(C)	21	(A)	27	(B)	33	(A)	39	(A)		
4	(C)	10	(B)	16	(B)	22	(A)	28	(B)	34	(C)	40	(A)		
5	(C)	11	(B)	17	(A)	23	(C)	29	(D)	35	(A)	41	(B)		
6	(B)	12	(B)	18	(A)	24	(A)	30	(C)	36	(A)				

COMPREHENSIVE CONCEPTUAL SOLUTIONS & RATIONALES:

Q.1 — Official Correct Answer: (A) 1 — 2 — 3 — 4

Question: Arrange the following left-bank tributaries of the River Ganga from West to East in sequential order: 1. Gomti 2. Ghaghara 3. Gandak 4. Kosi Select the correct answer using the code given below:
Solution Rationale: From west to east, the major northern Himalayan tributaries joining the Ganga are: Ramganga, Gomti, Ghaghara, Gandak, and Kosi.

Q.2 — Official Correct Answer: (C) Russia

Question: BrahMos is a supersonic cruise missile developed as a joint venture between India and:
Solution Rationale: BrahMos (named after the Brahmaputra and Moskva rivers) is a joint venture between India's Defence Research and Development Organisation (DRDO) and Russia's NPO Mashinostroyeniya.

Q.3 — Official Correct Answer: (C) Both 1 and 2

Question: Consider the following statements regarding the Inter-Tropical Convergence Zone (ITCZ): 1. In the month of July, the ITCZ shifts northward over the Indo-Gangetic Plain (monsoon trough). 2. The ITCZ is a broad trough of low pressure characterized by convective clouds and heavy rainfall. Which of the statements given above is/are correct?
Solution Rationale: Both statements are correct: The ITCZ moves northwards towards the Tropic of Cancer in boreal summer, positioning over northern India where it is referred to as the monsoon trough.

Q.4 — Official Correct Answer: (C) Both 1 and 2

Question: Consider the following statements regarding Iodine in human physiology: 1. Iodine is an essential mineral required by the thyroid gland to synthesize thyroxine hormone. 2. Deficiency of iodine in diet leads to simple goiter. Which of the statements given above is/are correct?
Solution Rationale: Both statements are correct: Iodine is incorporated into thyroxine (T4) and triiodothyronine (T3). Inadequate dietary iodine causes thyroid enlargement (goiter).

Q.5 — Official Correct Answer: (C) Mining and Quarrying

Question: During the year 2009-2010, which one of the following sectors recorded the highest growth rate in the Indian economy?
Solution Rationale: According to Economic Survey data for 2009-10, the manufacturing and industrial sectors recorded double-digit expansion following stimulus measures.

Q.6 — Official Correct Answer: (B) Rocky

Question: Extrusive volcanoes are not found in
Solution Rationale: Official UPSC CDS Solution: Assessed under Geography (Physical, Indian & World) according to UPSC CDS standards.

Q.7 — Official Correct Answer: (A) Nashik Road (Maharashtra)

Question: In which of the following cities is the India Security Press (ISP), which manufactures passports, postage stamps, and judicial stamp papers, located?
Solution Rationale: The India Security Press (ISP) is located at Nashik Road, Maharashtra, under SPMCIL. Bank notes are printed at Dewas and Nashik.

Q.8 — Official Correct Answer: (C) Barren Island

Question: In which one of the following islands of India is an active volcano found?
Solution Rationale: Barren Island in the Andaman and Nicobar Islands is the only confirmed active volcano in India and the South Asian subcontinent.

Q.9 — Official Correct Answer: (A) Dehradun

Question: In which one of the following cities is the headquarters of the Forest Survey of India (FSI) located?
Solution Rationale: The Forest Survey of India (FSI), an organization under the Ministry of Environment, Forest and Climate Change engaged in national forest cover mapping and assessments, is headquartered in Dehradun, Uttarakhand.

Q.10 — Official Correct Answer: (B) 1, 2 and 3 only

Question: Which of the following National Parks of India are declared as World Heritage Sites by UNESCO? 1. Keoladeo National Park 2. Sundarbans National Park 3. Kaziranga National Park 4. Ranthambore National Park Select the correct answer using the code given below :
Solution Rationale: Keoladeo National Park (1985), Kaziranga National Park (1985), and Sundarbans National Park (1987) are UNESCO World Heritage Sites. Ranthambore National Park is not inscribed on the World Heritage List.

Q.11 — Official Correct Answer: (B) Southern Hemisphere

Question: Roaring Forties, Furious Fifties, and Shrieking Sixties are terms applied to strong westerly winds found
Solution Rationale: In the Southern Hemisphere between 40° and 60°S latitude, the lack of landmasses allows the prevailing westerlies to blow with extreme unobstructed velocity, termed Roaring Forties, Furious Fifties, and Shrieking Sixties.

Q.12 — Official Correct Answer: (B) Himachal Pradesh

Question: The Rohtang Pass, a high mountain pass connecting the Kullu Valley with Lahaul and Spiti, is situated in which Indian state?
Solution Rationale: Rohtang Pass (elevation 3,978 m) is situated on the eastern Pir Panjal Range in Himachal Pradesh, connecting the Kullu Valley with the Lahaul and Spiti Valleys.

Q.13 — Official Correct Answer: (A) Brahmaputra and Moskva

Question: The supersonic cruise missile 'BrahMos' developed through a joint venture between India and Russia is named after which two rivers?
Solution Rationale: BrahMos is named as a portmanteau of the Brahmaputra River in India and the Moskva River in Russia.

Q.14 — Official Correct Answer: (C) Odisha

Question: Through which of the following states does the Tropic of Cancer (23°30' N) NOT pass in India?
Solution Rationale: The Tropic of Cancer passes through 8 Indian states: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, and Mizoram. It does NOT pass through Odisha.

Q.15 — Official Correct Answer: (C) Palk Strait

Question: What is the name of the strait where 'Adams' bridge' is situated ?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Geography (Physical, Indian & World)).

Q.16 — Official Correct Answer: (B) A rift valley (graben)

Question: What is the primary characteristic geological feature of the Narmada River Valley?
Solution Rationale: The Narmada River flows westward through a linear rift valley (fault graben) bounded between the Vindhya Range to the north and the Satpura Range to the south.

Q.17 — Official Correct Answer: (A) Demonstration of the end of Great Britain and France as dominant imperial superpowers

Question: What was the major international consequence of the Suez Crisis of 1956 following Egypt's nationalization of the canal?
Solution Rationale: The Suez Crisis forced Britain and France to withdraw under US and UN pressure, accelerating global decolonization and highlighting their displacement by the US and USSR as premier superpowers.

Q.18 — Official Correct Answer: (A) Western Ghats

Question: Which among the following regions of India is officially designated as a Global Biodiversity Hotspot?
Solution Rationale: India contains 4 recognized global biodiversity hotspots: The Himalayas, Indo-Burma, Western Ghats & Sri Lanka, and Sundaland (Nicobar Islands).

Q.19 — Official Correct Answer: (A) Mahanadi — Godavari — Krishna — Cauvery

Question: Which is the correct arrangement of the following peninsular rivers of India when arranged from North to South?
Solution Rationale: From north to south: Mahanadi (Chhattisgarh/Odisha) -> Godavari (Maharashtra/AP) -> Krishna (Maharashtra/Karnataka/AP) -> Cauvery (Karnataka/Tamil Nadu).

Q.20 — Official Correct Answer: (A) Barail Range

Question: Which mountain hill range forms the natural physical boundary separating the state of Manipur from the state of Nagaland?
Solution Rationale: The Barail Range is a tertiary mountain chain that extends across Assam, Nagaland, and Manipur, forming the water divide and physical boundary between Nagaland and Manipur.

Q.21 — Official Correct Answer: (A) Damodar River

Question: Which of the following river basins is known as the 'Sorrow of Bengal'?
Solution Rationale: The Damodar River was historically known as the 'Sorrow of Bengal' due to frequent catastrophic floods before the construction of the Damodar Valley Corporation (DVC) dams. Kosi is the Sorrow of Bihar.

Q.22 — Official Correct Answer: (A) Calcium oxide (Quicklime, CaO)

Question: Which of the following chemical substances dissolves in water with a vigorous evolution of heat (strongly exothermic reaction)?
Solution Rationale: Reaction of quicklime with water forms slaked lime [CaO + H2O -> Ca(OH)2] and releases large amounts of thermal energy (slaking of lime).

Q.23 — Official Correct Answer: (C) Blood

Question: Which one among the following animal tissués transports hormones and heat and maintains water balance ?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Geography (Physical, Indian & World)).

Q.24 — Official Correct Answer: (A) Volcanic islands

Question: Which one among the following is the geological origin of the Comoros archipelago situated in the northern Mozambique Channel of the Indian Ocean?
Solution Rationale: The Comoros islands are volcanic in origin, formed over an oceanic hot spot. Mount Karthala on Grande Comore is an active shield volcano.

Q.25 — Official Correct Answer: (A) India Against Corruption (IAC)

Question: Which one among the following was the nationwide anti-corruption citizen movement launched in 2011 advocating the enactment of the Jan Lokpal Bill?
Solution Rationale: The 2011 Indian anti-corruption movement, organized primarily under the banner of India Against Corruption (IAC) and spearheaded by Anna Hazare, demanded an independent Lokpal.

Q.26 — Official Correct Answer: (A) Rain water

Question: Which one among the following is the purest natural source of fresh water in the earth's hydrological cycle before touching ground pollutants?
Solution Rationale: Rainwater is the purest form of natural water as it is produced by the natural evaporation-condensation cycle (distillation process) in the atmosphere.

Q.27 — Official Correct Answer: (B) Aravalli Range

Question: Which one of the following is the oldest mountain range in India?
Solution Rationale: The Aravalli Range is one of the oldest fold mountain systems in the world, formed during the Proterozoic eon, now deeply eroded into relict mountains.

Q.28 — Official Correct Answer: (B) Benguela Current

Question: Which one of the following ocean currents is a cold current
Solution Rationale: The Benguela Current is a cold ocean current flowing northward along the southwest coast of Africa in the South Atlantic Ocean. Gulf Stream and Brazil Current are warm Atlantic currents; Kuroshio is in the Pacific.

Q.29 — Official Correct Answer: (D) Sambhar Salt Lake

Question: Which one of the following inland lakes in India has the highest water salinity?
Solution Rationale: Sambhar Salt Lake in Rajasthan is India's largest inland salt lake and has the highest salinity among Indian inland lakes.

Q.30 — Official Correct Answer: (C) Madhya Pradesh

Question: Which one of the following States of India is the largest producer of lignite coal?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Geography (Physical, Indian & World)).

Q.31 — Official Correct Answer: (D) Australia

Question: Which one of the following countries is located entirely south of the Equator?
Solution Rationale: Australia lies entirely in the Southern Hemisphere, well south of the Equator (latitudes 10°S to 43°S). Colombia, Indonesia, and Kenya are all crossed directly by the Equator.

Q.32 — Official Correct Answer: (C) Orient Express

Question: Which one of the following trains/ railways passes through Germany, France, Austria, Hungary and Romania?
Solution Rationale: Official UPSC CDS Rationale: Tested in CDS General Knowledge (Geography (Physical, Indian & World)).

Q.33 — Official Correct Answer: (A) Rapids — Ox-bow lake — Estuary

Question: Which one of the following is the correct sequence of the topographical features of a river from its upper course to its lower course?
Solution Rationale: Rapids form in the steep upper (youthful) course; meanders and ox-bow lakes form on the flat floodplains of the middle/lower course; estuaries or deltas form where the river meets the sea at its mouth.

Q.34 — Official Correct Answer: (C) Vancomycin

Question: Which one of the following glycopeptide antibiotics is commonly reserved in clinical practice as the 'drug of last resort' against severe multi-drug resistant Gram-positive infections (such as MRSA)?
Solution Rationale: Vancomycin is a powerful glycopeptide antibiotic historically used as a drug of last resort for serious, life-threatening infections caused by Gram-positive bacteria unresponsive to other antibiotics.

Q.35 — Official Correct Answer: (A) Spring Tide

Question: Which one of the following ocean tides is produced when the Sun, the Moon, and the Earth are aligned in a straight line (syzygy)?
Solution Rationale: Spring tides occur twice each month (during new moon and full moon) when gravitational pulls of the Sun and Moon reinforce each other in a straight line, producing the highest high tides.

Q.36 — Official Correct Answer: (A) Gujarat

Question: Which one of the following Indian states has the longest coastline among all peninsular states of mainland India?
Solution Rationale: Gujarat has the longest mainland coastline in India at approximately 1,600 km, followed by Andhra Pradesh (approx. 974 km).

Q.37 — Official Correct Answer: (A) Benguela Current

Question: Which one of the following is a cold ocean current flowing equatorward along the southwestern coast of the African continent?
Solution Rationale: The Benguela Current is a prominent cold eastern boundary current of the South Atlantic subtropical gyre flowing north along the coast of Namibia and South Africa.

Q.38 — Official Correct Answer: (A) Mediterranean Climate

Question: Which one of the following global climate types is characterized by mild, wet winters and warm, dry summers, supporting extensive citrus fruit orchards and vineyards?
Solution Rationale: The Mediterranean climate (Cs in Köppen classification) features dry summers under subtropical high pressure and rainy winters dominated by mid-latitude westerlies, ideal for olives and grapes.

Q.39 — Official Correct Answer: (A) Tapti (Tapi)

Question: Which one of the following major rivers of India flows westward and empties into the Arabian Sea rather than the Bay of Bengal?
Solution Rationale: The Narmada and Tapti rivers flow westward through tectonic rift valleys and empty into the Arabian Sea via the Gulf of Khambhat. Mahanadi, Godavari, and Cauvery flow eastward into the Bay of Bengal.

Q.40 — Official Correct Answer: (A) Tundra Biome

Question: Which type of natural biome is characterized by low temperatures, short growing seasons, permafrost, and dominant vegetation consisting of mosses and lichens?
Solution Rationale: The Tundra biome is treeless and circumpolar in high northern latitudes. Due to subsoil permafrost and biting winds, only low-lying mosses, lichens, and dwarf shrubs survive.

Q.41 — Official Correct Answer: (B) Gopal Krishna Gokhale

Question: Who among the following was the founder of the 'Servants of India Society' in 1905?
Solution Rationale: Gopal Krishna Gokhale founded the Servants of India Society in Pune in 1905 to train national missionaries for the service of India and promote social advancement.